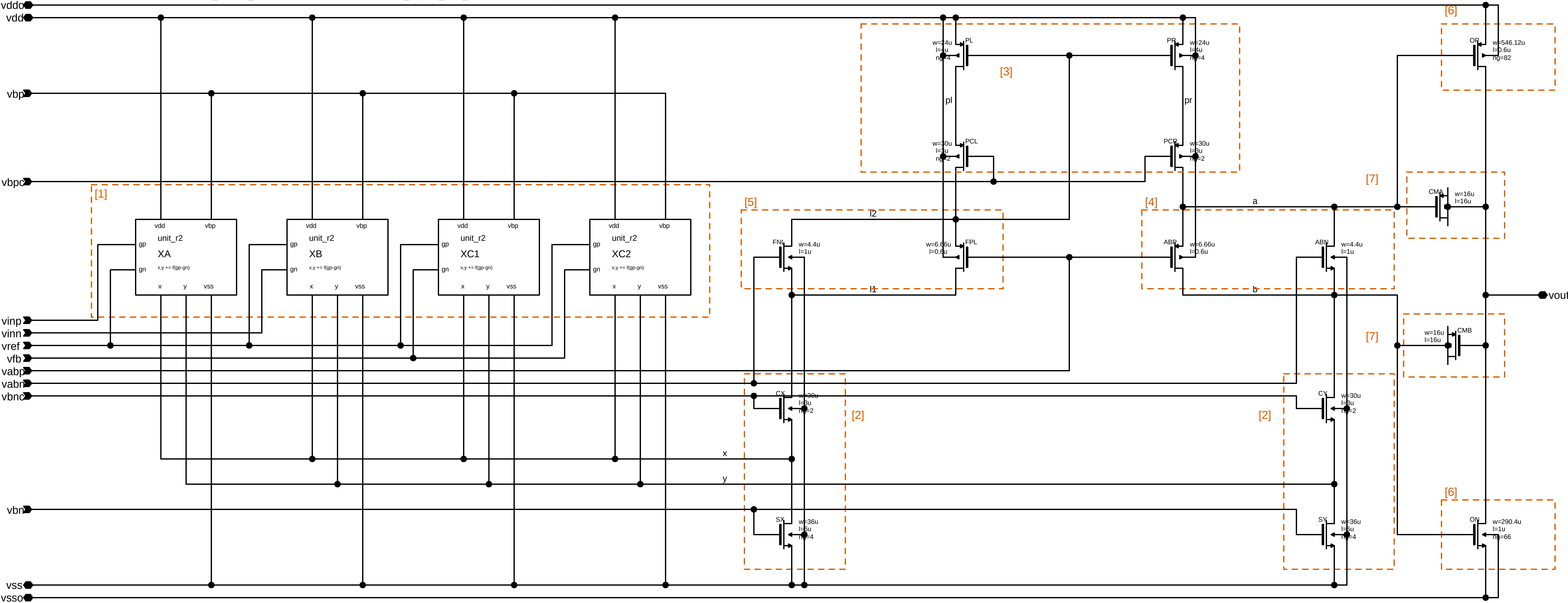


d2s_mpdda: matched-pair DDA (4 x unit_r2), folded cascode, class-AB output, MOS Miller compensation (vfb = vout)

frozen .subckt defaults: lcas = 3 (CX, CY, PCL, PCR 30u / 3u ng=2), wc = lc = 16u (CMA, CMB), rl = 50.6u (in unit_r2.sch)
the DDA units at transistor level: d2s_mpdda_flat.sch; with the bias network: d2s_mpdda_bias_flat.sch; block descriptions: README.md



[1] DDA units A, B, C1, C2: split-tail PMOS pair, rhigh between the sources. Each adds $f(gp - gn)$ to the fold nodes: the gp device drains to y, the gn device to x.

A: $vinp - vref$, B: $vref - vinn$, C1 = C2: $vref - vfb$. The loop forces $f(vinp - vref) + f(vref - vinn) = 2 f(vfb - vref)$, so $vfb - vref = (vinp - vinn)/2$.

[2] fold: sinks SX, SY (gate vbn) and NMOS cascodes CX, CY (gate vbnc). x and y are the fold nodes.

[3] PMOS cascoded mirror: PL, PCL is the input side (gates tied to I2), PR, PCR the output side into node a. Cascode gates vbpc.

[4] class-AB control: ABP (gate vabp) and ABN (gate vabn) between a and b. With RP1/RP2 and RN1/RN2 in the bias it sets the quiescent current of OP and ON.

[5] FPL, FNL: copy of [4] in the mirror input branch (between I2 and I1), so both fold branches carry the same element.

[6] output devices OP (gate a), ON (gate b), drains on vout.

[7] Miller capacitors CMA (between a and vout) and CMB (between vout and b): hv PMOS in accumulation, gate over n-well, 16u x 16u.